

Recap on Relations and properties of Relations.

A binary relation R on sets A, B is a subset of $A \times B$

$$R \subseteq A \times B$$

Properties of relation R defined on a single set.

$$R \subseteq A \times A$$

- Reflexive

$$\forall a \in A : a R a$$

- Irreflexive

$$\forall a \in A : a \not R a$$

$$A = \{1, 2\}$$



This is neither reflexive nor irreflexive.

- Symmetric

$$\forall a_1, a_2 \in A : a_1 R a_2 \Rightarrow a_2 R a_1$$

If you have a forward edge you need to have a backward edge.



Symmetric



not-symmetric



Symmetric

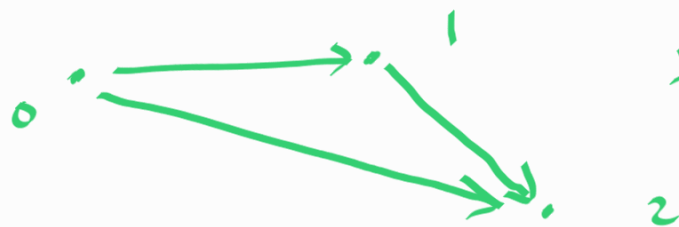
Anti-symmetric

$$\forall a_1, a_2 \in A : (a_1 R a_2) \wedge (a_2 R a_1) \Rightarrow a_1 = a_2$$



Transitive

$$\forall a_1, a_2, a_3 \in A: (a_1 R a_2) \wedge (a_2 R a_3) \Rightarrow a_1 R a_3$$



transitive



transitive

$a_1 = 0$	$a_1 = 0$
$a_2 = 1$	$a_2 = 2$
$a_3 = 2$	$a_3 = 1$

$(0 \rightarrow 1)$ ✓	$(1 \rightarrow 2)$ ✓
$a_1 R a_2 \wedge a_2 R a_3$	✓
✓	✓



not transitive

$a_1 = 0$	$(0 \rightarrow 1)$ ✓	$(1 \rightarrow 2)$ ✓
$a_2 = 1$	$a_1 R a_2 \wedge a_2 R a_3$	✓
$a_3 = 2$	✓	

$$A = \{0, 1, 2\}$$

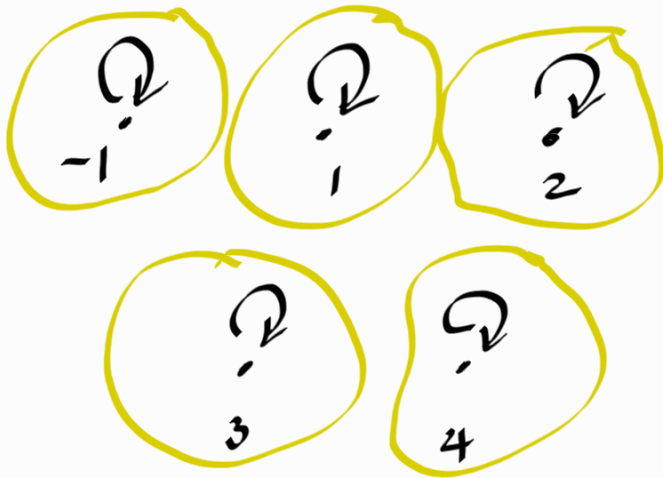
$$R = \{ \langle a, b \rangle \in A \times A : a \mid b \}$$



$a_1 = 1$	$a_1 R a_2$ ✓
$a_2 = 2$	$a_2 R a_3$ ✓
$a_3 = 0$	$a_1 R a_3$

Def Relation R on Set A , is an equivalence relation, if R is reflexive, symmetric and transitive.

$$A = \{-1, 1, 2, 3, 4\}$$



Reflexive ✓
Symmetric ✓
Transitive ✓

Equivalence relation

$$R_1 = \{ \langle -1, 1 \rangle, \langle 1, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 3 \rangle, \langle 4, 4 \rangle \}$$

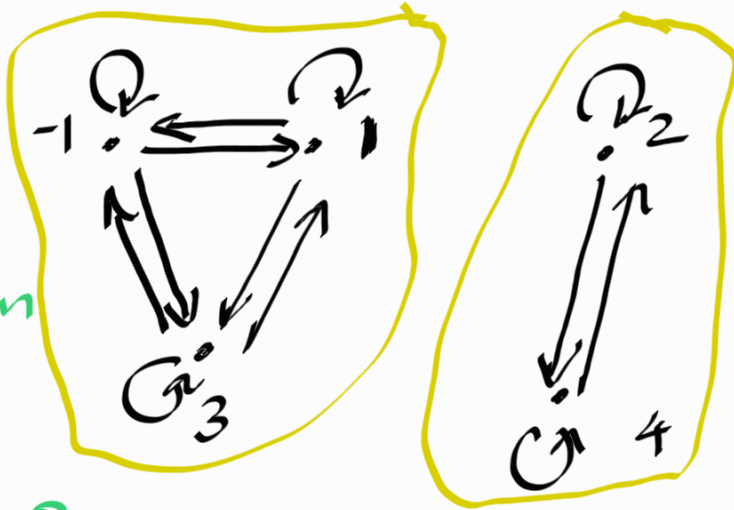
$$R_1 \subseteq A \times A$$

$$[-1] = \{-1\} \quad [1] = \{1\} \quad [2] = \{2\}$$

$$A = \{-1, 1, 2, 3, 4\}$$

$$[3] = \{3, 1, -1\} = [-1] = [1]$$

Same
parity
relation



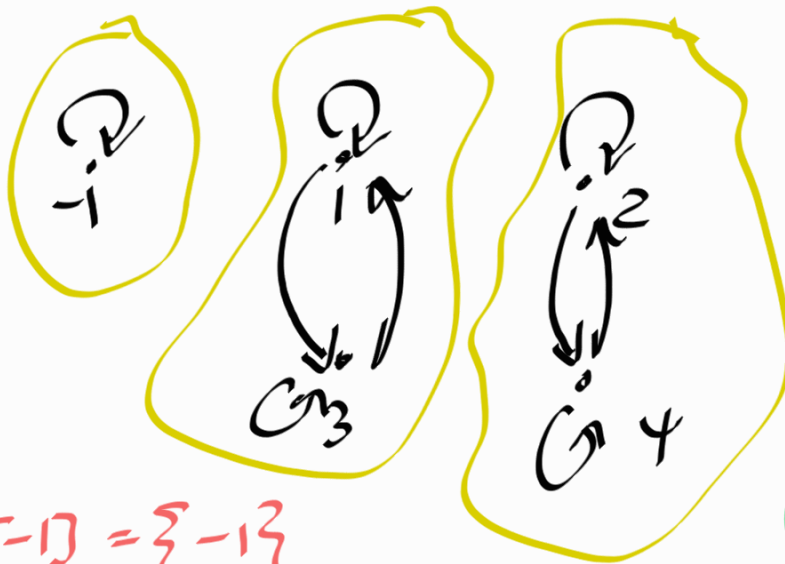
Reflexive ✓
Symmetric ✓
Transitive ✓

Equivalence
Relation

$$R_2 = \{ \langle -1, -1 \rangle, \langle 1, 1 \rangle, \langle 3, 3 \rangle, \langle -1, 1 \rangle, \langle 1, -1 \rangle, \langle 1, 3 \rangle, \langle 3, 1 \rangle, \langle -1, 3 \rangle, \langle 3, -1 \rangle, \langle 2, 2 \rangle, \langle 4, 4 \rangle, \langle 2, 4 \rangle, \langle 4, 2 \rangle \}$$

$$[2] = \{2, 4\}$$

Same
sign
and
same
parity



Reflexive ✓
Symmetric ✓
Transitive ✓

Equivalence
Relation

$$[-1] = \{-1\}$$

$$[1] = \{1, 3\}$$

$$[3] = \{3, 1\} = [1] \quad [2] = \{2, 4\}$$

Def: Equivalence classes

For an equivalence relation R on set A
the equivalence class of $a \in A$ is:

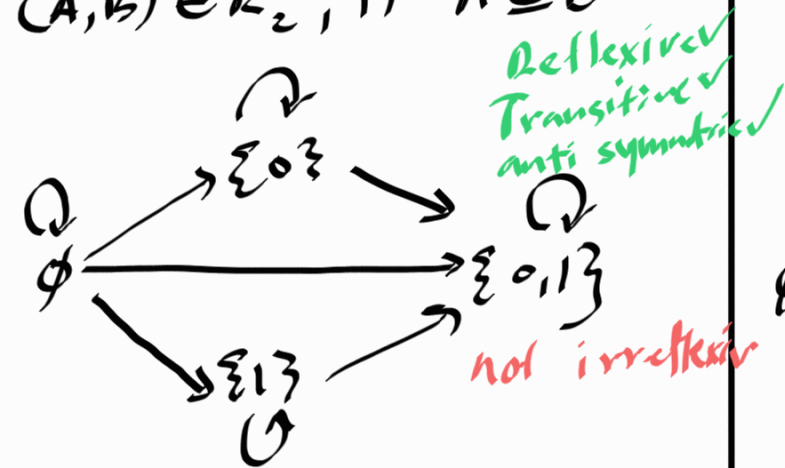
$$[a] = \{x \in A : x R a\}$$

set $S = P(\{0,1\}) = \{\{0\}, \{1\}, \{0,1\}, \emptyset\}$

$\subseteq$ relation R_2

$R_2 \subseteq S \times S$

$(A,B) \in R_2$, if $A \subseteq B$



$(\emptyset, \emptyset) \in R_2$ $\emptyset \subseteq \emptyset$

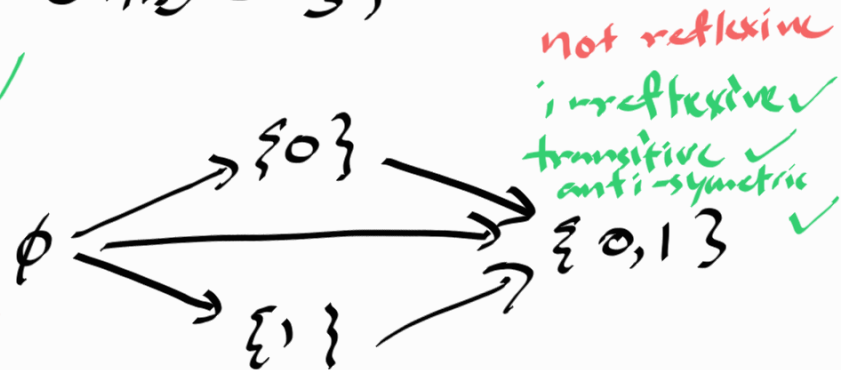
$(\{0\}, \{0\}) \in R_2$ $\{0\} \subseteq \{0\}$

$(\{0\}, \{1\}) \notin R_2$ $\{0\} \not\subseteq \{1\}$

$\subset$ relation R_3

$R_3 \subseteq S \times S$

$(A,B) \in R_3$, if $A \subset B$



$(\emptyset, \emptyset) \notin R_3$ $\emptyset \not\subset \emptyset$

$(\{1\}, \{1\}) \notin R_3$

$(\emptyset, \{0\}) \in R_3$ $\emptyset \subset \{0\}$

$(\{0\}, \{0,1\}) \in R_3$ $\{0\} \subset \{0,1\}$

Def: A Binary relation R on set A is:

1. A partial order, If R is

- Reflexive

- transitive

- anti-symmetric

R_2 is a partial order

R_3 is not a partial order

2. A strict partial order, if R is

- irreflexive

- transitive

- anti-symmetric

R_2 is not a strict partial order

R_3 is a strict partial order

Def

① A partial order is a total order if all pairs of different elements from A are comparable.

$$\forall a_1, a_2 \in A : (a_1 \neq a_2) \Rightarrow (a_1 R a_2) \vee (a_2 R a_1)$$

② A strict partial order a strict total order, if all pairs of different elements from A are comparable

$$\forall a_1, a_2 \in A : (a_1 \neq a_2) \Rightarrow (a_1 R a_2) \vee (a_2 R a_1)$$

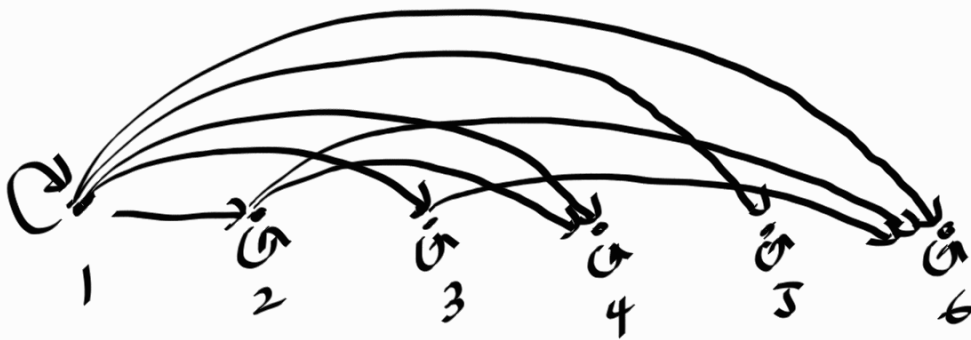
Popur Quiz 07

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$R = \{(a, b) \in S \times S : a \mid b\}$$

$$R = \{(1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (1, 1), (2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (5, 5), (6, 6)\}$$

b)



$$2. \quad A = \{0, 1, 2, 3, 4\} \quad B = \{0, 1, 2, 3\}$$

a) $a = b$

$$R = \{(0, 0), (1, 1), (2, 2), (3, 3)\}$$

b) $a + b = 4$

$$R = \{(1, 3), (2, 2), (3, 1), (4, 0)\}$$

c) $R = \{(1, 0), (2, 0), (2, 1), (3, 0), (3, 1), (3, 2), (4, 0)\}$

$$(4,1), (4,2), (4,3) \}$$